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Experiment 1

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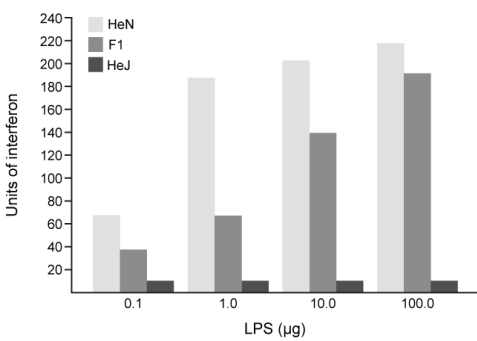


Figure 1 Average serum interferon levels in mice cohorts

Experiment 3

Mice were injected with varying doses of Ph-LPS and monitored for 4 days. The lethal dose (LD₅₀) was defined as the minimum dose causing death in 50% of injected mice. Compared to HeN mice, results reflected a 10-fold increase in LD₅₀ for F1 mice and a 100-fold increase in LD₅₀ for HeJ mice.

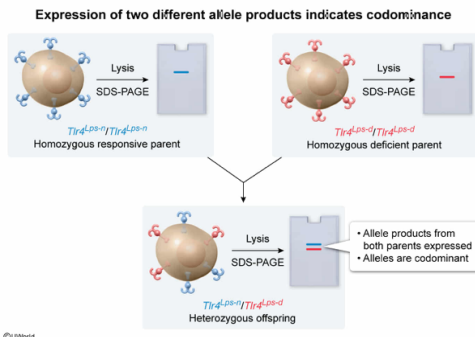
Experimental results suggest that the mutated *Tlr4* allele exhibits which quality?

- ✓ ☒ A. Codominance [34%]
- ☐ B. Recessivity [21%]
- ☐ C. Reduced penetrance [15%]
- ☐ D. Variable expressivity [27%]

Correct

34% answered correctly

Explanation:



Codominance describes the **co-expression of alleles** as observed in the phenotype of heterozygous individuals. In the case of codominant traits, neither allele is fully dominant or fully recessive. A common example of codominance is the co-expression of both red (R) and white (W) alleles in a speckled offspring (RW) in some flowers.

In the passage, *in vitro* studies of **F1** spleen cells (ie, spleen cells from the offspring of a **cross** between the parental HeN and HeJ mice, resulting in TLR heterozygotes) suggest that **both wild-type and mutant TLRs** are observable by protein electrophoresis. This co-expression of the normally functioning receptor and the mutated receptor can be attributed to the **codominance of $tlr4^{Lps-n}$ and $tlr4^{Lps-d}$** alleles. In the presence of Ph-LPS, F1 cells [$tlr4^{Lps-n}/tlr4^{Lps-d}$] display intermediate production of anti-SRC antibodies and interferon because half of their expressed TLR receptors are functional while the other half of expressed proteins are mutant. As a result, the **quantitative data** for their immune response were lower than HeN mice [$tlr4^{Lps-n}/tlr4^{Lps-n}$] mice but higher than HeJ mice [$tlr4^{Lps-d}/tlr4^{Lps-d}$].

(Choice B) If the mutant allele were recessive, F1 mice (which are heterozygous; [$tlr4^{Lps-n}/tlr4^{Lps-d}$]) would exhibit a full response to LPS, and measurements observed in the F1 generation would be

indistinguishable from measurements observed in the HeN mice ($tlr4^{Lps-n}/tlr4^{Lps-n}$).

(Choice C) **Penetrance** refers to the *portion* of offspring that expresses a genetic trait. If the $tlr4^{Lps-d}/tlr4^{Lps-d}$ mutation had reduced penetrance, only a fraction of the F1 offspring would show decreased response to LPS compared to HeN mice. As seen in Table 1, immune response was decreased relatively uniformly among F1 mice as indicated by the low **standard error**.

(Choice D) **Expressivity** refers to the *range of symptoms* observed in individuals with a given genetic condition. If the $tlr4^{Lps-d}$ allele had variable expressivity, the mutation would result in a range of different phenotype characteristics. As seen in Table 1, immune response was decreased relatively uniformly among F1 mice, as indicated by the low standard error.

Educational objective:

Codominance refers to the co-expression of two alleles at the same locus, where both allelic products are observable in the phenotype of heterozygous individuals.

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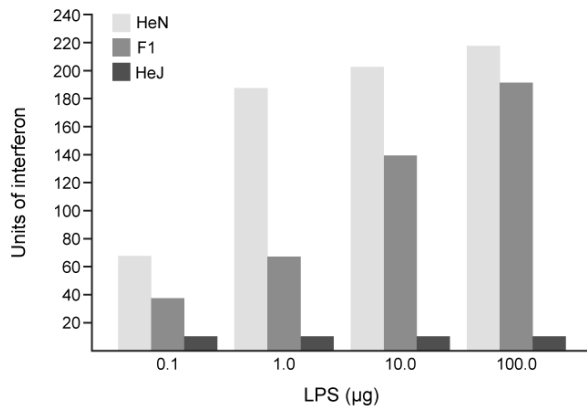


Figure 1 Average serum interferon levels in mice cohorts

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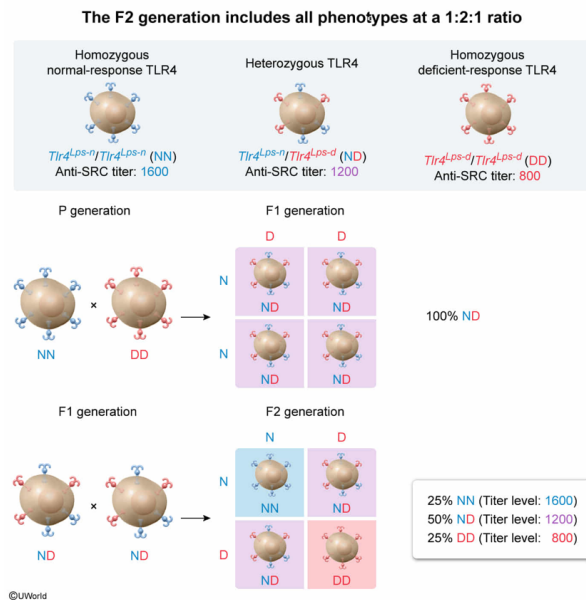
If researchers repeated the analyses using F2 generation mice, spleen cell cultures exposed to SRC plus 1.0 µg Ph-LPS would be expected to have anti-SRC titers of:

- I. 1600
 - II. 1200
 - III. 800
- ☐ A. II only [17%]
 - ☐ B. I and II only [17%]
 - ☐ C. I and III only [10%]
 - ☒ D. I, II, and III [54%]

Correct

53% answered correctly

Explanation:



The experiment describes a **monohybrid cross**, or a mating between individuals with different alleles at one genetic locus. The first cross is performed between homozygous HeN ($tlr4^{Lps-n}/tlr4^{Lps-n}$) and HeJ mice ($tlr4^{Lps-d}/tlr4^{Lps-d}$), members of the parental or P generation, and produces a heterozygous F1 generation ($tlr4^{Lps-n}/tlr4^{Lps-d}$). The second cross would be performed between members of the F1 generation and would produce an **F2 generation** with a **mix of genotypes**. Depending on the genotype, some F2 offspring would be *phenotypically similar* to members of the P generation, whereas some F2 offspring would resemble members of the F1 generation.

For instance, expected F2 genotypes can be used to deduce anti-SRC titers when spleen cell cultures are exposed to SRC plus 1.0 μ g Ph-LPS:

- $tlr4^{Lps-n}/tlr4^{Lps-n}$: An estimated 25% of F2 mice would be homozygous for the normal functioning *Tlr4* allele. These mice would have an immune response similar to that seen in HeN mice. According to Table 1, this corresponds to an antibody titer of 1605 ± 55 (**Number I**).
- $tlr4^{Lps-n}/tlr4^{Lps-d}$: An estimated 50% of mice would be heterozygous for the *Tlr4* gene mutation. These mice would have an immune response similar to F1 generation mice, or an antibody titer of 1240 ± 58 (**Number II**).
- $tlr4^{Lps-d}/tlr4^{Lps-d}$: An estimated 25% of mice would be homozygous for the *Tlr4* gene mutation. These mice would have an immune response similar to HeJ mice, or an antibody titer of 810 ± 17 (**Number III**).

Educational objective:

In a monohybrid cross, members of the P generation are homozygous for different alleles at one locus. The first cross produces heterozygous offspring in the F1 generation. The second cross produces a mix of genotypes and phenotypes observed in members of the F2 generation.

Time spent:0 sec

Subject:

Biology

QID:400209

System:

Genetics and Evolution

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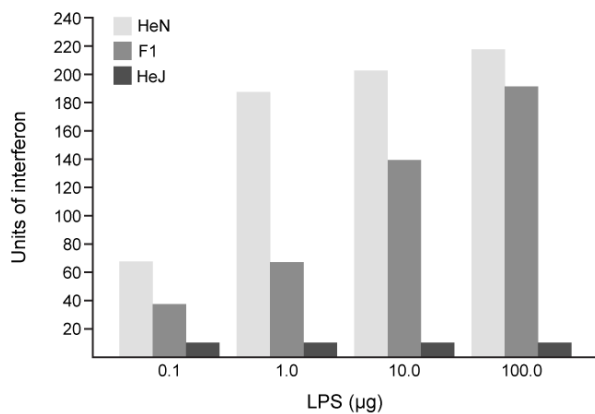


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In an isolated population of 10,000 mice, 1,600 are homozygous for a *Tlr4* defect. Assuming stable allele and genotype frequencies, how many mice are heterozygous for a *Tlr4* defect?

- ☐ A. 2,400 [11%]
- ☐ B. 3,600 [12%]
- ☒ C. 4,800 [62%]
- ☐ D. 8,400 [12%]

Correct

61% answered correctly

Explanation:

Hardy-Weinberg equilibrium

1. Determine alleles & possible genotypes

Alleles	Possible genotypes
N (wild-type responsive $Tlr4^{Lps-n}$)	NN Homozygous responsive
D (mutant defective $Tlr4^{Lps-d}$)	ND Heterozygous
	DD Homozygous defective

2. Use HWE equations to derive frequency of ND

Allele frequency using HWE equation 1	Genotype frequency using HWE equation 2
$p + q = 1$	$p^2 + 2pq + q^2 = 1$
p = frequency of N	p^2 = frequency of NN
q = frequency of D	$2pq$ = frequency of ND
	q^2 = frequency of DD

3. Use given information to calculate q^2

1. Given that $q^2 = 0.16$ solve for q using $p + q = 1$
2. Use $2pq$ to calculate the frequency of ND
3. $2pq = 2(0.6)(0.4) = 0.48$
4. 48% of 10,000 = 4,800

HWE = Hardy-Weinberg equilibrium.

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Allele frequency describes the relative frequency of an allele or gene version within a given population, and **genotype frequency** reflects the relative frequency of a particular genotype. Usually, these frequencies are described as *fractions* or *percentages*. **Hardy-Weinberg calculations** can be used to relate allele and genotype frequencies in **stable, nonevolving populations**:

- **$p + q = 1$** : If two versions (alleles) exist for a given gene, respective allele frequencies are commonly denoted as p , which represents the major (most common) allele, and q , which represents the minor (less common) allele. Statistically, the sum of these frequencies must equal 1.
- **$p^2 + 2pq + q^2 = 1$** : Genotype frequency can be determined by the *probability* of inheriting a genotype and is therefore dependent on **allele frequencies**. The frequency of **homozygotes** is equal to p^2 or q^2 , whereas the frequency of **heterozygotes** is equal to $2pq$.

In this example, p is the frequency of the normal-functioning $Tlr4$ allele and q is the frequency of the mutated $Tlr4$ allele. The frequency of mice homozygous for the mutated allele (q^2) is 1,600/10,000, or 0.16. Therefore, q is the square root of 0.16, or 0.40, and this value can be used to calculate p :

$$p = 1 - 0.40 = 0.60$$

The p and q allele frequencies can then be used to calculate the frequency of the heterozygous genotype and the total number of heterozygous mice in the population of 10,000:

$$2pq = 2(0.60)(0.40) = 0.48$$

$$(0.48)(10,000) = 4,800 \text{ mice}$$

(Choice A) 2,400 is one-half of the number of heterozygous mice. Heterozygote frequency is equal to *2 times pq*, which is the sum of probabilities of inheriting *p* from the father and *q* from the mother (*1pq*) or *p* from the mother and *q* from the father (*1pq*).

(Choice B) 3,600 is equal to the number of mice homozygous for the normal functioning receptor: $(10,000)(p^2)$.

(Choice D) 8,400 is equal to the number of mice containing at least one copy of the normal allele: $(10,000)(p^2 + 2pq)$ or $(10,000)(1 - q^2)$.

Educational objective:

Hardy-Weinberg equations can be used to relate allele frequencies and genotype frequencies. Genotype frequencies are equal to the probability of inheriting each genotype, with the frequency of homozygotes equal to p^2 or q^2 and the frequency of heterozygotes equal to $2pq$.

Time spent:0 sec

Subject:
Biology

QID:400210

System:
Genetics and Evolution

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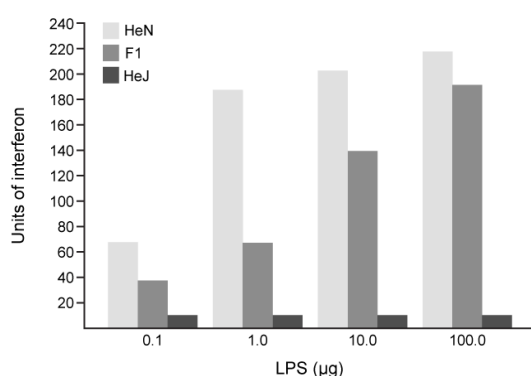


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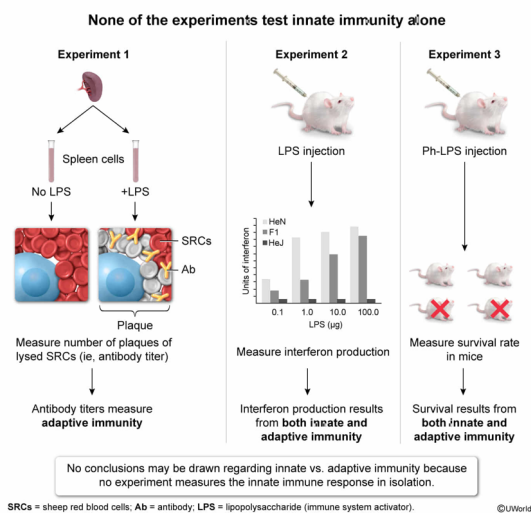
The experiments allowed researchers to determine the effect of genotype variation on all of the following EXCEPT:

- ☐ A. *in vitro* versus *in vivo* studies. [18%]
- ☒ B. innate versus adaptive immune responses. [32%]
- ☐ C. responses modulated by LPS exposure. [1%]
- ☐ D. qualitative and quantitative phenotypic traits. [47%]

Correct

32% answered correctly

Explanation:



The researchers in the passage performed several experiments to examine the effect of *Tlr4* genotype on mice. Experiment 1 was performed on spleen cells isolated from mice of various strains. Because these experiments were performed outside a living organism, these experiments are considered *in vitro* experiments. The spleen cells from different genotypic strains were *qualitatively* characterized based on the types of proteins expressed via protein electrophoresis, and then *quantitatively* characterized based on antibody titer (Choice D).

Experiments 2 and 3 were performed in living mice and were therefore *in vivo* experiments. Combined with Experiment 1, these studies provide a comparison for *in vitro* effects in Experiment 1 versus *in vivo* effects in Experiments 2 and 3 (Choice A). In all three experiments, the mice or mouse cells were exposed to varying amounts of LPS for observing the resulting response (Choice C).

All three experiments also characterized the immune response, both *innate* and *adaptive*. However, none of the experiments can be used *to compare* innate versus adaptive immune responses. While Experiment 1 measures antibody levels (specifically adaptive immunity), both the results of Experiments 2 and 3 measure *combined effects* of both innate *and* adaptive

immunity. **Interferons**, measured in Experiment 2, represent a large class of signaling molecules that act in **both** the **innate and adaptive** immune systems; and survival, measured in Experiment 3, is an outcome of *overstimulation* of both the innate and adaptive immune responses. Because *none of the experiments are specific to innate immunity alone*, researchers would not be able to differentiate the influence of genotype on innate *versus* adaptive immune response.

Educational objective:

Experiments must be designed to isolate effects of the variables being tested. To compare innate vs adaptive immunity, innate immunity in isolation must be tested against adaptive immunity in isolation. In living organisms, both the innate and adaptive immune responses generally work together to protect the organism against foreign antigens.

Time spent:0 sec

Subject:
Biology

QID:400211

System:
Genetics and Evolution

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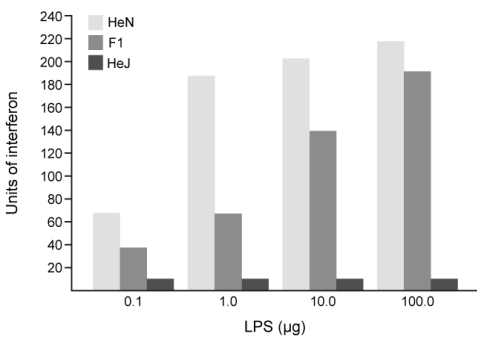
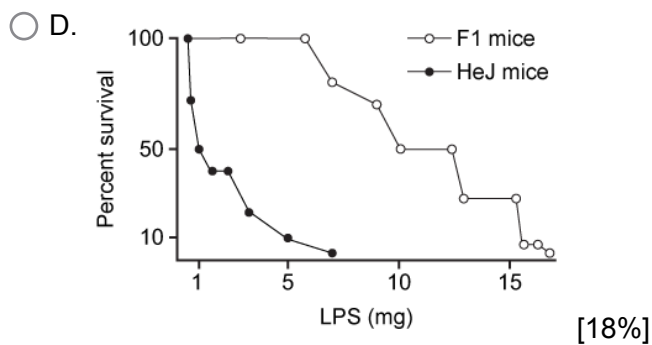
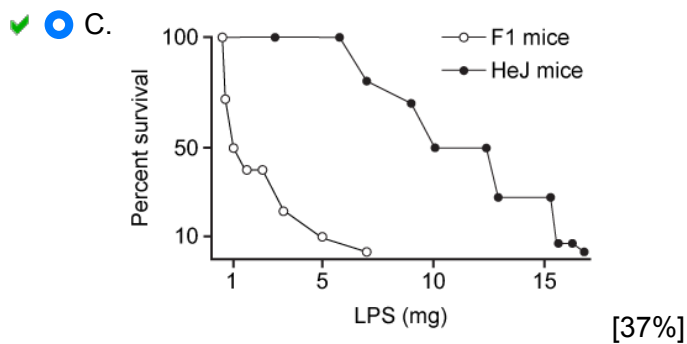
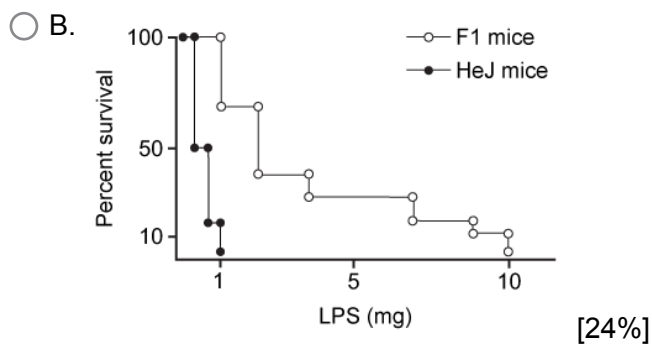
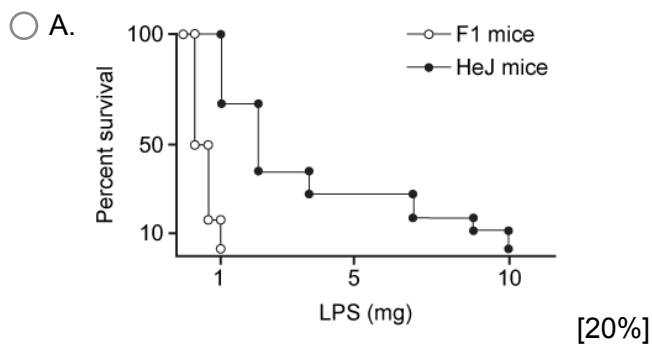


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According to the results of Experiment 3, which of the following could represent survival curves for F1 and HeJ mice?



Correct

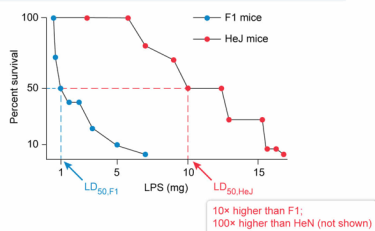
36% answered correctly

Explanation:

HeJ mice have a 10× higher LD₅₀ than F1 mice

From passage:
 $LD_{50,F1} = 10 \times LD_{50,HeN}$
 $LD_{50,HeJ} = 100 \times LD_{50,HeN}$

Calculate:
 $\frac{LD_{50,HeJ}}{LD_{50,F1}} = \frac{100 \times LD_{50,HeN}}{10 \times LD_{50,HeN}} = \frac{10}{1} \rightarrow LD_{50,HeJ} = 10 \times LD_{50,F1}$



Survival curves plot the fraction or number of surviving members as a function of an independent variable,

微信Mai_ange

usually time after diagnosis (for humans) or drug dosage. In animal-based safety studies, survival curves are often used to assess the **lethal dose** value LD_{50} , which is the **minimum dose** causing **death in 50%** of group members. On survival curves, this data point can be observed as the dose (x-value) corresponding to 50% survival (y-value).

In this case, survival is a function of LPS dose. Experiment 3 states that compared to the LD_{50} for *HeN* mice, 10-fold and 100-fold increases in LD_{50} were found for F1 and HeJ mice, respectively. This result means that the LD_{50} for HeJ mice was **10 times greater** than the LD_{50} for F1 mice. The only survival curve that corresponds to this ratio is the one that shows that 50% of F1 mice died at a dose of 1 mg ($LD_{50} = 1$ mg) and that 50% of HeJ mice died at a 10× higher dose of 10 mg ($LD_{50} = 10$ mg).

(Choices A and B) These curves correspond to LD_{50} values near 0.5 mg and 2.5 mg, or only a 5-fold difference.

(Choice D) In this curve, the LD_{50} for the F1 mice is 10 times greater than the LD_{50} for the HeJ mice. This is the opposite of what was observed in Experiment 3: LD_{50} is greater in HeJ mice, not F1 mice.

Educational objective:

Survival curves plot percent survival against an independent variable, usually time or dosage. Survival curves can be used to determine LD_{50} , defined as the minimum dose causing death in 50% of test subjects.

Time spent:0 sec
Subject:
Biology

QID:400212
System:
Genetics and Evolution

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Experiment 1

Researchers isolated spleen cells from both parents as well as the progeny of the cross for analysis of protein expression and LPS response. In cells isolated from mice from the F1 generation, both wild-type and mutant TLRs were distinguishable by protein electrophoresis. Following this first analysis, spleen cells from each cohort (HeN mice, HeJ mice, or F1 progeny) were exposed to 2×10^5 sheep erythrocytes (SRCs), which served as a source of an antigens to stimulate the mouse immune cells. The spleen cell suspension was then incubated in the presence or absence of Ph-LPS (LPS extracted from *Escherichia coli* K345 cells using a phenol–water mixture). Titers (antibody levels) of anti-SRC antibody per spleen were recorded on incubation day 5, with the results shown in Table 1. Titers were measured as the number of plaques of lysed erythrocytes surrounding an activated immune cell.

Table 1 Mean Anti-SRC Titers with Standard Error

Spleen cell source	Anti-SRC titer levels		
	0 µg Ph-LPS	0.1 µg Ph-LPS	1.0 µg Ph-LPS
HeN	840 ± 46	1235 ± 95	1605 ± 55
HeJ	767 ± 45	703 ± 41	810 ± 17
F1	880 ± 35	907 ± 38	1240 ± 58

Experiment 2

Serum samples were assayed for interferon production 2 hours post–LPS injection. Interferon was quantified using a plaque reduction assay on mouse cells challenged with vesicular stomatitis virus.

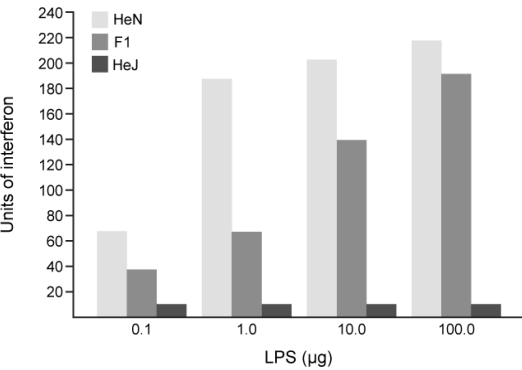


Figure 1 Average serum interferon levels in mice cohorts

Experiment 3

Mice were injected with varying doses of Ph-LPS and monitored for 4 days. The lethal dose (LD₅₀) was defined as the minimum dose causing death in 50% of injected mice. Compared to HeN mice, results reflected a 10-fold increase in LD₅₀ for F1 mice and a 100-fold increase in LD₅₀ for HeJ mice.

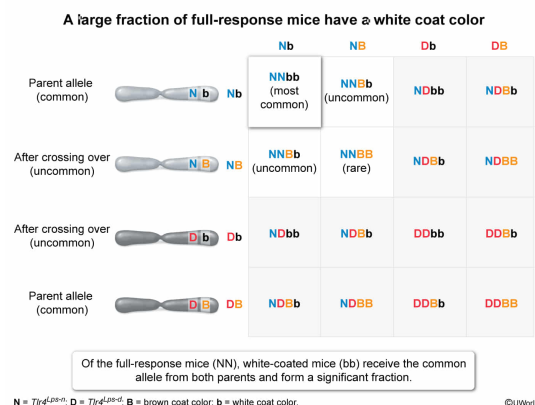
In mice, a brown coat color is a dominant trait. Assuming that the HeN mice in the experiments were homozygous for the white allele and the HeJ mice were homozygous for the brown allele, which of the following is true with regard to coat color and LPS response in subsequent generations?

- ☐ A. F1 mice will produce gametes with equal haplotype frequencies. [31%]
- ✓ ☒ B. A large fraction of F2 mice exhibiting full response to LPS will be white. [35%]
- ☐ C. A large fraction of F1 mice exhibiting full response to LPS will be white. [18%]
- ☐ D. Half of F1 mice exhibiting full response to LPS will be brown and the other half will be white. [14%]

Correct

35% answered correctly

Explanation:



For alleles on different chromosomes, the **law of independent assortment** states that when the alleles separate during **meiosis**, they will do so **independently** of each other. In other words, there will be equal haplotype frequencies for alleles *on different chromosomes*. However, the passage states that the *Tlr* gene and the coat color gene are close together on the **same chromosome** (Chromosome 4). Therefore, these genes are **linked** rather than independent.

Genetic linkage refers to the tendency of alleles **on the same chromosome** to **remain on the same chromosome**. Linked alleles will *not* separate independently during meiosis unless a **crossing-over event** happens. As the genes become closer in proximity, the frequency of

crossing over between them decreases, and the genes are said to be **tightly or closely linked**. In these cases, the various haplotypes will *not* be produced at equal frequencies and a **linkage disequilibrium** is said to occur.

The F1 generation, formed by the cross of white (*w/w*) HeN mice and homozygous brown (*B/B*) HeJ mice, have a genotype of *w, Tlr4^{Lps-n}/B, Tlr4^{Lps-d}*. Because coat color and the *Tlr4* gene are closely linked, the gametes will *not* have equal haplotype frequencies (**Choice A**). Instead, coat

color and *Tlr4* will tend to stay together, and the majority of the gametes produced by F1 mice will have chromosomes with the same, **nonrecombinant** haplotypes they inherited from the parental generation (ie, $w, Tlr4^{Lps-n}$ and $B, Tlr4^{Lps-d}$); only a small fraction of gametes will show recombinant haplotypes (ie, $B, Tlr4^{Lps-n}$ and $w, Tlr4^{Lps-d}$).

Therefore, when these F1 mice mate with each other, offspring that are fully responsive to LPS will have a *Tlr* genotype of $Tlr4^{Lps-n}/Tlr4^{Lps-n}$, and because the white coat color allele is linked to the responsive *Tlr* allele, the majority of these mice will have a full genotype of $w, Tlr4^{Lps-n}/w, Tlr4^{Lps-n}$ and be **both fully responsive to LPS and white**.

(Choices C and D) Because the brown allele is dominant, *none* of the heterozygous F1 mice will be white (ie, for white coat color, mice must have two recessive *w* alleles). In addition, because the two *Tlr* alleles are codominant, *none* of the heterozygous F1 mice will show a full response to LPS.

Educational objective:

Genetic linkage refers to the tendency of alleles in close proximity to remain on the same chromosome and be inherited together by offspring. This tendency occurs because of fewer crossover events between these loci during meiosis, resulting in a greater number of haploid gametes with nonrecombinant genotypes.

Time spent:0 sec

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